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Java SE 27 & JDK 27

Interface Checksum

All Known Implementing Classes:

Adler32, CRC32, CRC32C

`public interface Checksum`

An interface representing a data checksum.

Since:

1.1

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
`long`	`getValue()`	Returns the current checksum value.
`void`	`reset()`	Resets the checksum to its initial value.
`default void`	`update(byte[] b)`	Updates the current checksum with the specified array of bytes.
`void`	`update(byte[] b, int off, int len)`	Updates the current checksum with the specified array of bytes.
`void`	`update(int b)`	Updates the current checksum with the specified byte.
`default void`	`update (ByteBuffer buffer)`	Updates the current checksum with the bytes from the specified buffer.

Method Details

update

`void update(int b)`

Updates the current checksum with the specified byte.

Parameters:

`b` - the byte to update the checksum with

update

```
default void update(byte[] b)
```

Updates the current checksum with the specified array of bytes.

Implementation Requirements:

This default implementation is equal to calling `update(b, 0, b.length)`.

Parameters:

`b` - the array of bytes to update the checksum with

Throws:

`NullPointerException` - if `b` is `null`

Since:

9

update

```
void update(byte[] b,
            int off,
            int len)
```

Updates the current checksum with the specified array of bytes.

Parameters:

`b` - the byte array to update the checksum with

`off` - the start offset of the data

`len` - the number of bytes to use for the update

update

```
default void update(ByteBuffer buffer)
```

Updates the current checksum with the bytes from the specified buffer. The checksum is updated with the remaining bytes in the buffer, starting at the buffer's position. Upon return, the buffer's position will be updated to its limit; its limit will not have been changed.

API Note:

For best performance with DirectByteBuffer and other ByteBuffer implementations without a backing array implementers of this interface should override this method.

Implementation Requirements:

The default implementation has the following behavior.
For ByteBuffers backed by an accessible byte array.

```
update(buffer.array(),
       buffer.position() + buffer.arrayOffset(),
       buffer.remaining());
```

For ByteBuffers not backed by an accessible byte array.

```
byte[] b = new byte[Math.min(buffer.remaining(), 4096)];
while (buffer.hasRemaining()) {
    int length = Math.min(buffer.remaining(), b.length);
    buffer.get(b, 0, length);
    update(b, 0, length);
}
```

Parameters:

`buffer` - the ByteBuffer to update the checksum with

Throws:

`NullPointerException` - if `buffer` is `null`

Since:

9

getValue

`long getValue()`

Returns the current checksum value.

Returns:

the current checksum value

reset

`void reset()`

Resets the checksum to its initial value.

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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